

Superstring amplitudes from the field-theory limit

Lecheng Ren

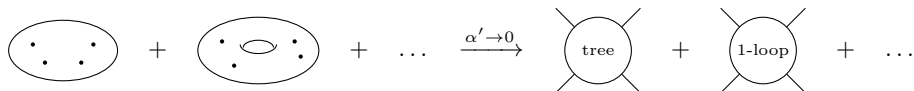
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based on: 2510.02455 with Ricardo Monteiro,
2410.19663 with Yvonne Geyer, Jiachen Guo, Ricardo Monteiro

April 21, 2026

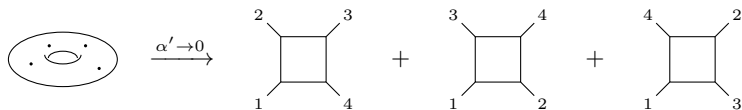
Focus on scattering of massless external states.

Field theory limit is $\alpha' \rightarrow 0$.



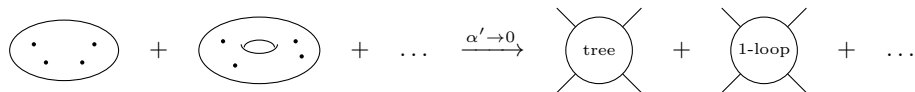
Powerful tool, e.g. first 1-loop results in $\mathcal{N} = 4$ SYM, $\mathcal{N} = 8$ SUGRA.

[Green, Schwarz, Brink 82']



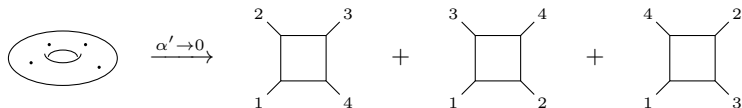
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Converse: from field theory back to string theory?

Factorisable external states, e.g. $\bar{\varepsilon}_{\mu\nu} = \varepsilon_\mu \tilde{\varepsilon}_\nu$.

[Mafra, Schlotterer, Stieberger 11']

$$\mathcal{A}_n^{(0)} = \alpha'^{n-2} \int_{\mathcal{M}_{0,n}} \frac{d^{2n}\sigma}{\text{vol SL}(2, \mathbb{C})} \left| \sum_{\rho \in S_{n-2}} \frac{\mathcal{N}(1, \rho(2), \dots, \rho(n-1), n)}{\sigma_{1\rho(2)} \sigma_{\rho(2)\rho(3)} \cdots \sigma_{\rho(n-1)n} \sigma_{n1}} \right|^2 \prod_{i < j} |\sigma_{ij}|^{\alpha' k_i \cdot k_j}$$

$$\sigma_{ij} = \sigma_i - \sigma_j \quad \overline{\mathcal{N}(\varepsilon_i)} \mapsto \mathcal{N}(\tilde{\varepsilon}_i)$$

- Simple dependence on stringy α' .
- Numerators $\mathcal{N}$ are field-theory “BCJ numerators”.

[Bern, Carrasco, Johansson 08']

$$\mathcal{A}_n^{(0)} = \sum_{\alpha' \in \text{cubic}} \frac{\mathcal{N}_{\alpha'}(\varepsilon_i) \tilde{\mathcal{N}}_\alpha(\tilde{\varepsilon}_i)}{\mathcal{D}_\alpha} + \mathcal{O}(\alpha'^3)$$

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Is (perturbative) string theory just an α' -dressed field theory?

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Is (perturbative) string theory just an α' -dressed field theory?

This talk: 1 loop any multiplicity (caveat for $n \geq 16$)

1-loop type II superstring amplitude

Chiral splitting [D'Hoker, Phong,...]

$$\mathcal{A}_n^{(1)} = \alpha'^n \int d^{10} \ell \int_{\mathcal{F}} d^2 \tau \int_{T^{n-1}} d^2 z_2 \dots d^2 z_n |\mathcal{I}_n|^2 |\text{KN}_n|^2$$

$$\text{KN}_n = \exp \frac{\alpha'}{2} \left(\sum_{1 \leq i < j \leq n} k_i \cdot k_j \ln \theta_1(z_{ij}, \tau) + 2i\pi \ell \cdot \sum_{j=1}^n z_j k_j + i\pi \tau \ell^2 \right)$$

Goal: construct chiral integrand $\mathcal{I}_n$

- modular weight: $n - 4$ [builds on Green, Schwarz, Brink; **Mafra, Schlotterer**, Stieberger; Cachazo, He, Yuan; Geyer, Mason, Monteiro; Tourkine, Vanhove; Adamo, Casali, Skinner; Mizera; Stark-Muchão]
- monodromy invariant: $\mathcal{I}_n|_{z_i \rightarrow z_i + \tau}^{l \rightarrow l - k_i} = \mathcal{I}_n$
- matches field theory ($\alpha' \rightarrow 0$): $\tau \rightarrow i\infty$



- Modular invariance: ansatz of weight $n - 4$

object	weight
$2\pi i \ell_\mu$	1
$g^{(w)}(z_i - z_j)$	w
$G_{2K}(\tau), (K > 1)$	$2K$

$$F(z, \alpha, \tau) := \frac{\theta'_1(0, \tau) \theta_1(z + \alpha, \tau)}{\theta_1(\alpha, \tau) \theta_1(z, \tau)} = \sum_{m=0}^{\infty} \alpha^{m-1} g^{(m)}(z, \tau), \quad G_{2K}(\tau) = -g^{(2K)}(0, \tau)$$

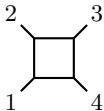
- Monodromy invariance: $\mathcal{I}_n|_{z_i \rightarrow z_i + \tau}^{\ell \rightarrow \ell - k_i} = \mathcal{I}_n$ ($g^{(w)}$ transform)
- Matches field theory ($\tau \rightarrow i\infty$):

$$\lim_{\tau \rightarrow i\infty} \mathcal{I}_n \prod_{i=2}^n dz_i = \frac{d^n \sigma}{\text{vol SL}(2, \mathbb{C})} \sum_{\rho \in S_n} \frac{N(\rho(1), \rho(2), \dots, \rho(n); \ell)}{\sigma_{+\rho(1)} \sigma_{\rho(1)\rho(2)} \sigma_{\rho(2)\rho(3)} \cdots \sigma_{\rho(n)-\sigma_{-+}}},$$

- $N(\dots; \ell)$ are n -gon BCJ numerators for 10D SYM

- $n = 4$ (weight 0)

ST from FT: $\mathcal{I}_4 = N(1, 2, 3, 4; \ell) = t_8(f_1, f_2, f_3, f_4)$, $f_i^{\mu\nu} = k_i^{[\mu} \varepsilon_i^{\nu]}$

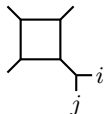


- $n = 5$ (weight 1)

ST: $\mathcal{I}_5 = 2\pi C_5^\mu \ell_\mu + \sum_{i < j} C_{5,ij} g_{ij}^{(1)}$

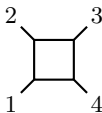
FT: $N(1, 2, 3, 4, 5; \ell) = C_5^\mu \ell_\mu - \frac{1}{2} \sum_{i < j} C_{5,ij}$ (from $\mathcal{I}_5|_{\tau \rightarrow i\infty}$)

ST from FT: $C_5^\mu = N(\dots)|_{\ell^1}$, $C_{5,ij} = -N(\dots[i, j]\dots)$



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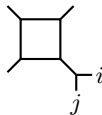


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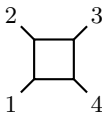
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Can get $\mathcal{I}_5$ from **BCJ numerators**.

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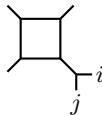


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ST from FT: $C_5^\mu = N(\dots)|_{\ell^1}$, $C_{5,ij} = -N(\dots[i, j]\dots)$



Can get $\mathcal{I}_5$ from **BCJ numerators**.

In fact, up to $\mathcal{I}_{15}$!

- $n = 5$ (weight 1)

$$\mathcal{I}_5 = N_5^\mu L_\mu - \sum_{2 \leq i < j \leq 5} N_5([i, j]) V_{1|i, j}, \quad N(1, 2, 3, 4, 5; \ell) = N_5^\mu \ell_\mu + \frac{1}{2} \sum_{i < j} N_5([i, j])$$

monodromy invariant basis:

$$L_\mu = 2\pi i \ell_\mu + \sum_{j=2}^n k_{j, \mu} g_{1, j}^{(1)}, \quad V_{1|i, j} := g_{1, i}^{(1)} + g_{i, j}^{(1)} - g_{1, j}^{(1)}$$

[Mafra, Schlotterer 17'-18']

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[Mafra, Schlotterer 17'-18']

- $n = 6$ (weight 2)

$$\mathcal{I}_6 = N_6^{\mu\nu} \mathbf{Reg}[L_\mu L_\nu] - \sum_{2 \leq i < j \leq 6} N_6^\mu([i, j]) \mathbf{Reg}[L_\mu V_{1|i, j}]$$

$$+ \sum_{\substack{2 \leq i < j, k \leq 6 \\ j \neq k}} N([i, j], [k]; \ell) V_{1|i, j, k} + \sum_{\substack{2 \leq i < j \leq 6 \\ 2 \leq k < l \leq 6, i < k}} N([i, j], [k, l]; \ell) V_{1|i, j} V_{1|k, l} \Big],$$

$\mathbf{Reg}[\dots]$ excludes closed cycles $g_{i_1 i_2}^{(w_1)} g_{i_2 i_3}^{(w_2)} \cdots g_{i_m i_1}^{(w_m)}$

- $n = 8$ (weight 4)

$$\begin{aligned} \mathcal{I}_8 = & \textcolor{red}{N}_8^{\mu\nu\rho\sigma} \mathbf{Reg}[L_\mu L_\nu L_\rho L_\sigma] - \sum_{2 \leq i < j \leq 7} \textcolor{violet}{N}_8^{\mu\nu\rho}([i, j]) \mathbf{Reg}[L_\mu L_\nu L_\rho V_1|_{i,j}] \\ & + \dots \text{ (lots of terms)} \\ & + \frac{(2\pi i)^4}{4!} \sum_{\rho \in S_7} \textcolor{red}{N}(1, \rho(2) \dots; \ell) \mathcal{E}_4(\tau) \end{aligned}$$

- Eisenstein series $\textcolor{violet}{G}_4(\tau)$ appears:
 - in its own right, conveniently normalised:

$$\textcolor{violet}{E}_{2k}(\tau) := \frac{G_{2k}(\tau)}{2\zeta(2k)} \xrightarrow{\tau \rightarrow i\infty} 1$$

- from regularisation $\mathbf{Reg}$

- $n = 8$ (weight 4)

$$\begin{aligned} \mathcal{I}_8 = & N_8^{\mu\nu\rho\sigma} \mathbf{Reg}[L_\mu L_\nu L_\rho L_\sigma] - \sum_{2 \leq i < j \leq 7} N_8^{\mu\nu\rho}([i, j]) \mathbf{Reg}[L_\mu L_\nu L_\rho V_{1|i, j}] \\ & + \dots \text{ (lots of terms)} \\ & + \frac{(2\pi i)^4}{4!} \sum_{\rho \in S_7} N(1, \rho(2) \dots; \ell) \mathcal{E}_4(\tau) \end{aligned}$$

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Coefficients of basis elements $\mathcal{E}_{2k}(\tau) \mathbf{Reg}[L \dots LV \dots V]$ fixed by **BCJ numerators**.

- $n = 16$ (weight 12)

$$\mathcal{I}_{16} = N_{16}^{\mu_1 \dots \mu_{12}} \mathbf{Reg}[L_{\mu_1} \dots L_{\mu_{12}}] + \dots \text{ (lots of terms)} \\ + C_{16, E_{12}} \mathcal{E}_{12}(\tau) + C_{16, \Delta} \Delta(\tau)$$

- $SL(2, \mathbb{Z})$ modular forms at weight 12 spanned by:

$$\mathcal{E}_{12}(\tau) := \frac{G_{12}(\tau)}{2\zeta(12)} \xrightarrow{\tau \rightarrow i\infty} 1, \quad \Delta(\tau) := \frac{(2\pi)^{12}}{12^3} [(\mathcal{E}_4(\tau))^3 - (\mathcal{E}_6(\tau))^2] \xrightarrow{\tau \rightarrow i\infty} 0$$

- Kinematic coefficients fixed by field-theory limit?

$$C_{16, \mathcal{E}_{12}} = \frac{(2\pi i)^{12}}{11!} \sum_{\rho \in S_{11}} N\left(1, \rho; -\frac{\rho_1}{2}\right)$$

- $C_{16, \Delta}$? Cusp basis elements **unfixed**.

- $n = 6$ (weight 2)

Recall

$$\mathcal{I}_6 = N_6^{\mu\nu} \mathbf{Reg}[L_\mu L_\nu] - \sum_{2 \leq i < j \leq 6} N_6^\mu([i, j]) \mathbf{Reg}[L_\mu V_{1|i, j}] + \dots$$

- Curious: piece $\sum_{\rho \in S_5} N(\rho; \ell)|_{\ell=0}$ unconstrained by BCJ relations, but follows from string theory.

New “ G_2 identities” in field theory ($n \geq 6$)

- $n = 6$ (weight 2)

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- $G_2(\tau)$ is quasi-modular: $G_{2K}(\frac{a\tau+b}{c\tau+d}) = (c\tau+d)^{2K} G_{2K}(\tau) - \frac{2\pi i c}{c\tau+d} \delta_{K,1}$
- Absence of $G_2(\tau)$ constrains BCJ numerators:

$$C_{6, G_2} = 0 \implies \frac{1}{6!} \sum_{\rho \in S_5} N(\rho; \ell)|_{\ell=0} = \frac{1}{12} N_6^{\mu\nu} \sum_{i=1}^6 k_{i, \mu} k_{i, \nu}$$

- Check on explicit 6-pt numerators. [Edison, He, Johansson, Schlotterer, Teng, Zhang 22’]
- Reduce number of undetermined coefficients in 7-pt numerators.
[EHJSTZ, 22’] [private comm. w/ He, Jing, Teng, Zhang]

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Loop integrand of maximal SYM/SUGRA (partially) constrained by **string modularity**.

- Constructed basis for 1-loop superstring correlators.
- Kinematic coefficients in this basis are
 - pieces of **BCJ numerators**, for non-cusp basis elements,
 - unfixed, for cusp basis elements ($n \geq 16$):
Do they vanish? Is string theory an α' -dressing of field theory?
- Higher loops? [D'Hoker, Schlotterer, 25']
- New “ G_2 constraints” on **BCJ numerators**: Field-theory interpretation?
- Double copy? [Stieberger's talk]

Thank you!